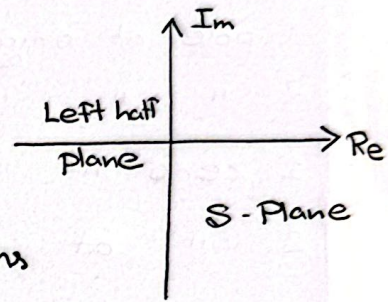


Bode Plot : It is the graphical representation of the T.F for determining the stability of the control system.

It is applicable to minimum phase T.F. In MPTF all the poles and zeros are in the left half of 'S' plane.

Let, a MPTF, $G(s)$ system, in which $R(t) = A \sin \omega t$ input is ~~given~~ and $A' \sin(\omega t + \phi)$ output is found. In output side A' amplitude variations happen and depending which we find gain and based on variation of phase, ϕ , we identify phase.



So, Bode plot consists of two separate plots. One plot is the logarithm of amplitude of a sinusoidal T.F. and the other is a plot of phase angle. Both are plotted against frequency.

The curves are drawn on semilog paper, using log scale for frequency and linear scale for amplitude (decibels) or phase angle (degrees).

The 2 different plots:

- i) Gain Plot $[|G(j\omega)|]$
- ii) Phase plot $[\angle G(j\omega)]$

Procedure of Bode Plot (Gain Plot)

Step 1: Write given TF in standard form.

$$\text{If } G(s) = \frac{(s+a)(s+b)}{(s+p)(s+q)} = \frac{ab(1+\frac{s}{a})(1+\frac{s}{b})}{pq(1+\frac{s}{p})(1+\frac{s}{q})}, \quad k = \frac{ab}{pq}$$

Step 2: Identify slope of 1st line, it is based on poles/zeros at the origin.

as the given system does not have any poles or zeros at the origin, so slope of 1st line is 0 dB/dec.

1 pole at origin, slope of 1st line = -20 dB/dec

2 " " " , " " " " = -40 dB/dec, and

1 zero " " " " " " = +20 dB/dec

2 " at " " " " " " = +40 dB/dec, and

no pole/zero at origin, slope of 1st line = 0 dB/dec.

Step 3: Identify Gain of 1st line at $\omega = 1$ rad/sec

$$\text{Gain}|_{\omega=1 \text{ rad/s}} = 20 \log k$$

Once this point and slope of 1st line is found, the 1st line of gain and other corner frequencies can be identified.

Step 4: Write all corner frequency in ascending order and define slope for each lines.

Corner F. $\rightarrow$ The frequency at which the slope of the Bode amplitude plot changes due to the presence of a pole or a zero.

$$\therefore a < p < b < q$$

$$G(s) = \frac{(s+a)(s+b)}{(s+p)(s+q)}$$

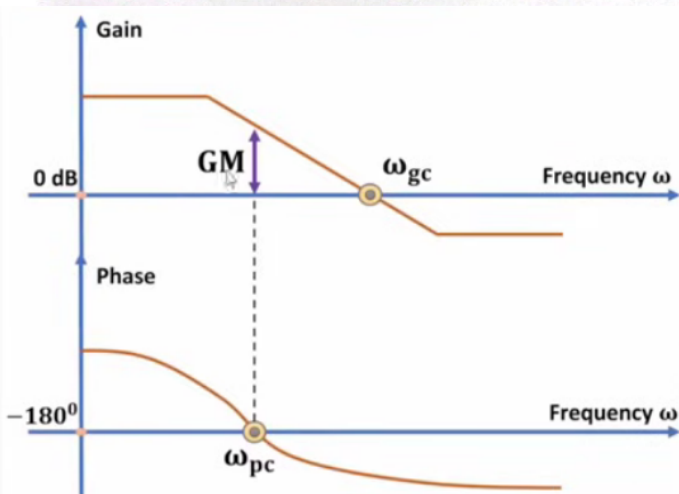
slope of 1st line 0 dB/dec as there is no pole/zero at the origin.

Corner F.	Pole/zero	slope	Resultant slope
a	zero	+20 dB/dec	= 20 dB/dec
p	pole	-20 dB/dec	= 0 dB/dec
b	zero	+20 dB/dec	= 20 dB/dec
q	pole	-20 dB/dec	= 0 dB/dec

Step 5: Phase Plot ($\angle G(j\omega)$)

$$G(j\omega) = \frac{(j\omega+a)(j\omega+b)}{(j\omega+p)(j\omega+q)} ; \text{ phase is positive for zeros and negative for poles.}$$

$$\angle G(j\omega) = \tan^{-1} \frac{\omega}{a} + \tan^{-1} \frac{\omega}{b} - \tan^{-1} \frac{\omega}{p} - \tan^{-1} \frac{\omega}{q}$$



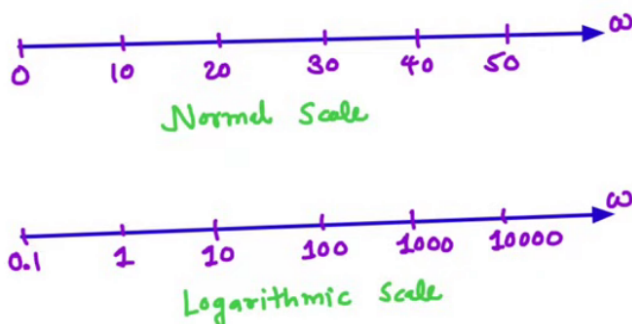
- ☐ **Gain Cross Over Frequency:** It is a frequency at which Gain Crosses 0 dB.
- ☐ **Phase Cross Over Frequency:** It is a frequency at which Phase Crosses -180° .
- ☐ **Gain Margin:** It is a margin of Gain with respect to 0 dB at Phase Cross Over Frequency.
 - Above 0 dB, Gain Margin is -Ve and Below 0 dB, Gain Margin is +Ve.
- ☐ **Phase Margin:** It is a margin of Phase with respect to -180° at Gain Cross Over Frequency.
 - Above -180° , Phase Margin is +Ve and Below -180° , Phase Margin is -Ve.

Stability using Bode Plot

- ☐ For Stable System: $\omega_{pc} > \omega_{gc}$
- ☐ For Stable System Gain Margin and Phase Margin is +Ve.

Advantages of Bode Plot

- ☐ Frequency Response Representation is Easy by Separate Magnitude and Phase.
- ☐ Logarithmic Scale Covers Broad Range of Frequency.



Q. Draw Bode plot for the TF: $G(s)H(s) = \frac{100(s+3)}{s(s+1)(s+5)}$

Also find: a) Gain cross over frequency, b) GM, c) PM)
d) Phase cross over frequency, e) Stability

Soln:
$$\frac{100 \times \frac{3}{5} (1 + \frac{s}{3})}{s(1+s)(1+\frac{s}{5})} = \frac{60 (1 + \frac{s}{3})}{s(1+s)(1+\frac{s}{5})}$$

Slope of 1st line = -20 dB/dec. as we have 1 pole at orig

Gain of 1st line at $\omega = 1$ rad/sec.

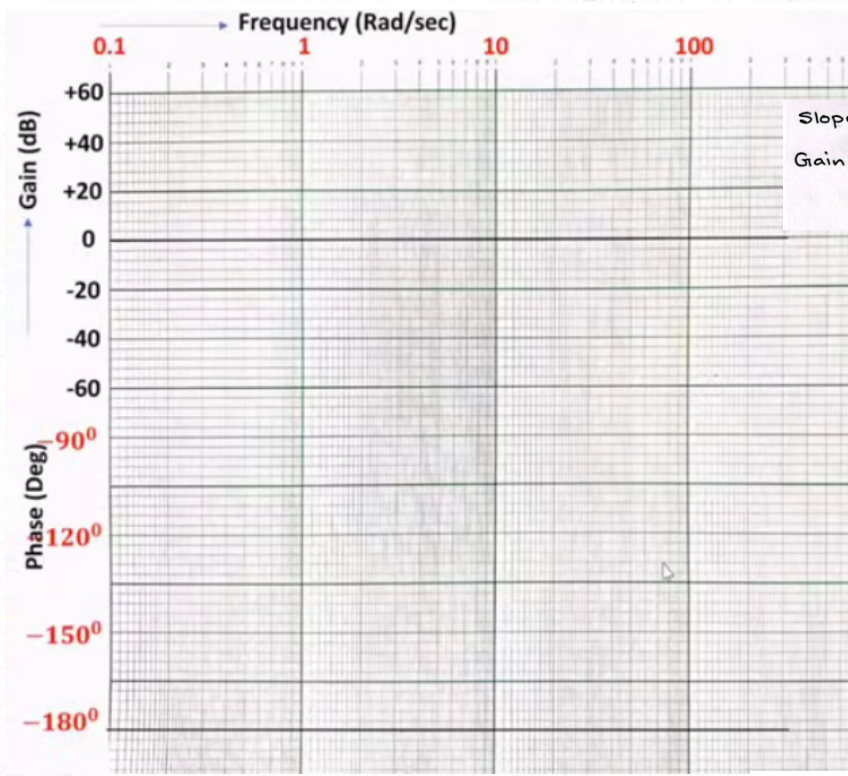
$$\text{Gain}|_{\omega=1 \text{ rad/s}} = 20 \log 60 = 35.5 \text{ dB.}$$

Corner F.	P/Z	slope	Resultant slope
0	P	-20 dB/dec	= -20 dB/dec
1	P	-20 dB/dec	= -40 dB/dec
3	Z	+20 dB/dec	= -20 dB/dec
5	P	-20 dB/dec	= -40 dB/dec

$$\angle G(j\omega) = -90 - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{5} + \tan^{-1}\frac{\omega}{3}$$

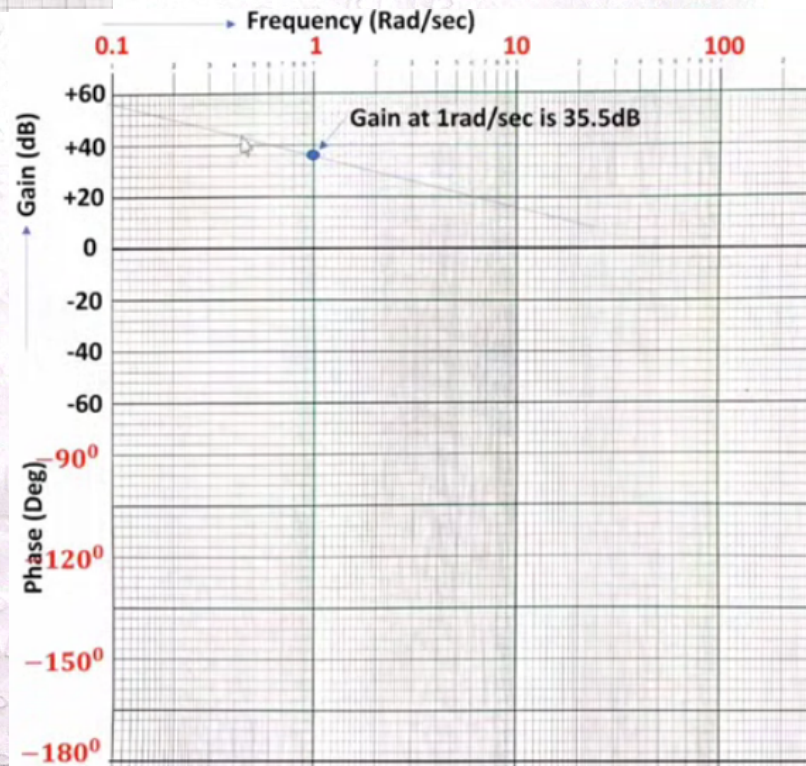
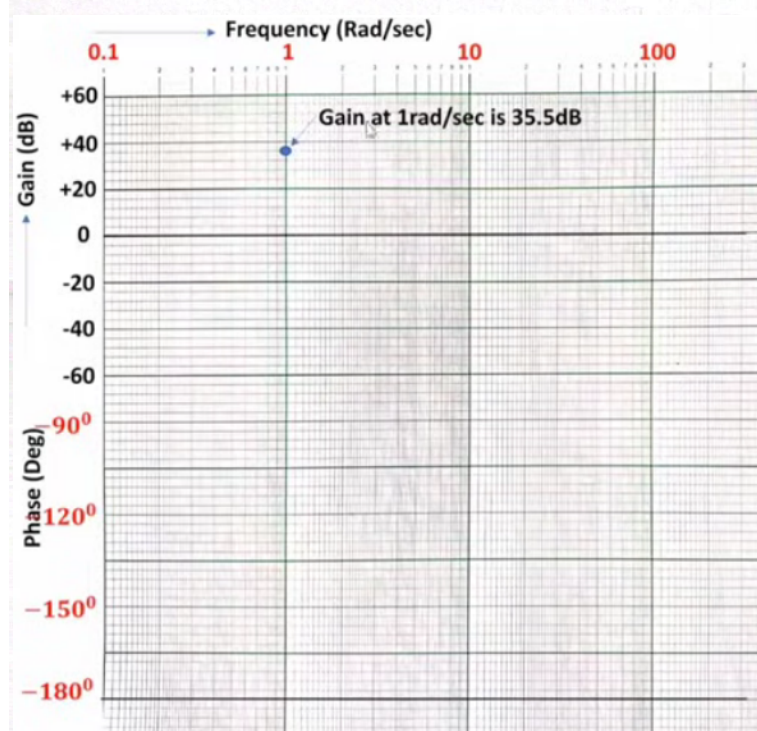
Make a table by taking some random frequencies.

$\omega = 0.1$ rad/s	-99.9°	$\omega = 10$ rad/s	-164.9°
$\omega = 0.5$ rad/s	-112.7°	$\omega = 100$ rad/s	-178.2
$\omega = 1$ rad/s	-127.9°	$\omega = 500$ rad/s	-179.6
$\omega = 5$ rad/s	-154.8°	$\omega = 1000$ rad/s	-179.8



Slope of 1st line = -20 dB/dec. as we have 1 pole at origin
 Gain of 1st line at $\omega = 1$ rad/sec.

$$\text{Gain}|_{\omega=1 \text{ rad/sec}} = 20 \log 60 = 35.5 \text{ dB.}$$



Connen F.

0
1
3
5

P/z

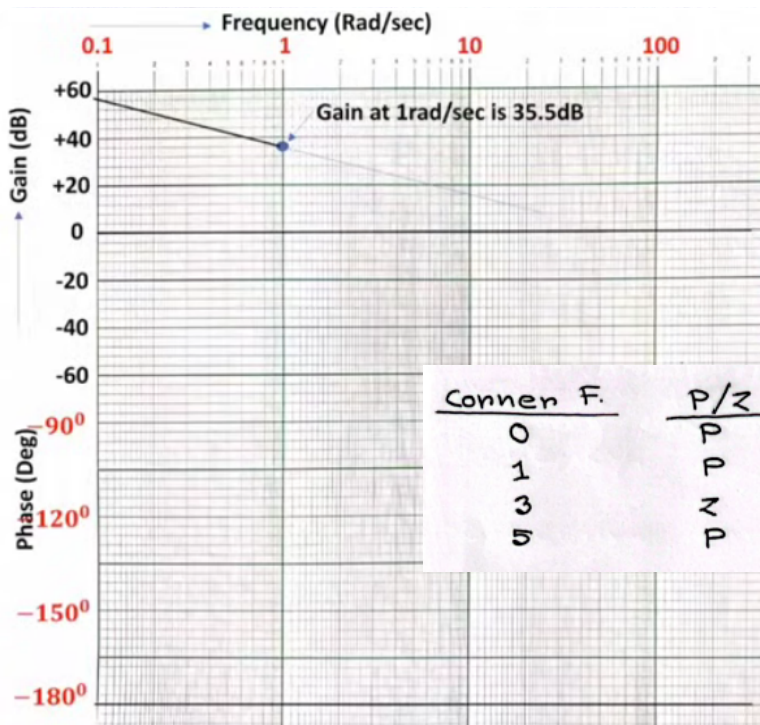
P
P
Z
P

slope

-20 dB/dec
-20 dB/dec
+20 dB/dec
-20 dB/dec

Resultant slope

= -20 dB/dec
= -40 dB/dec
= -20 dB/dec
= -40 dB/dec



Connen F.

P/Z

slope

Resultant slope

0

P

-20 dB/dec

= -20 dB/dec

1

P

-20 dB/dec

= -40 dB/dec

3

Z

+20 dB/dec

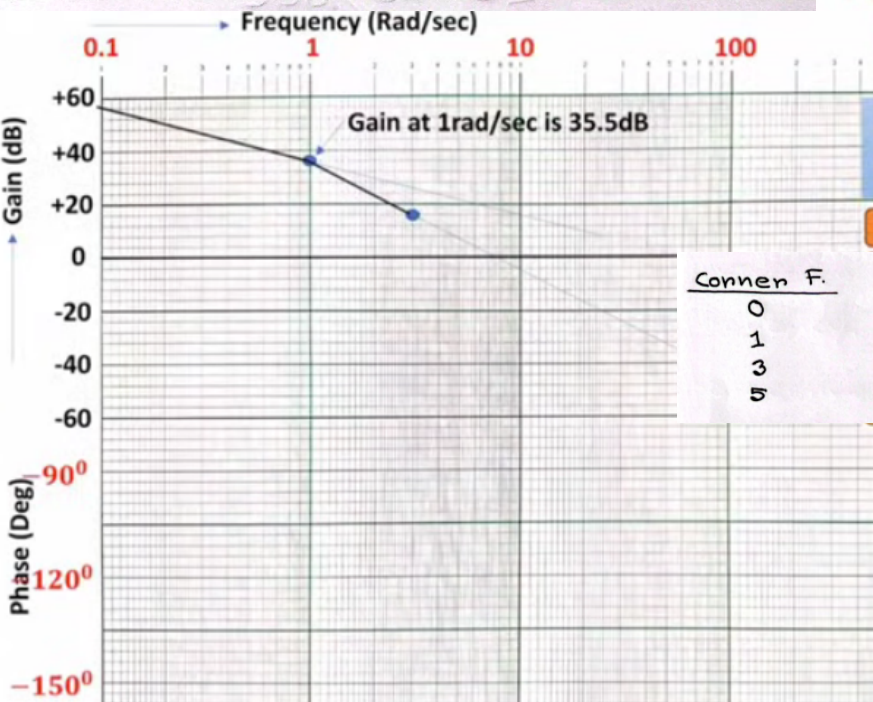
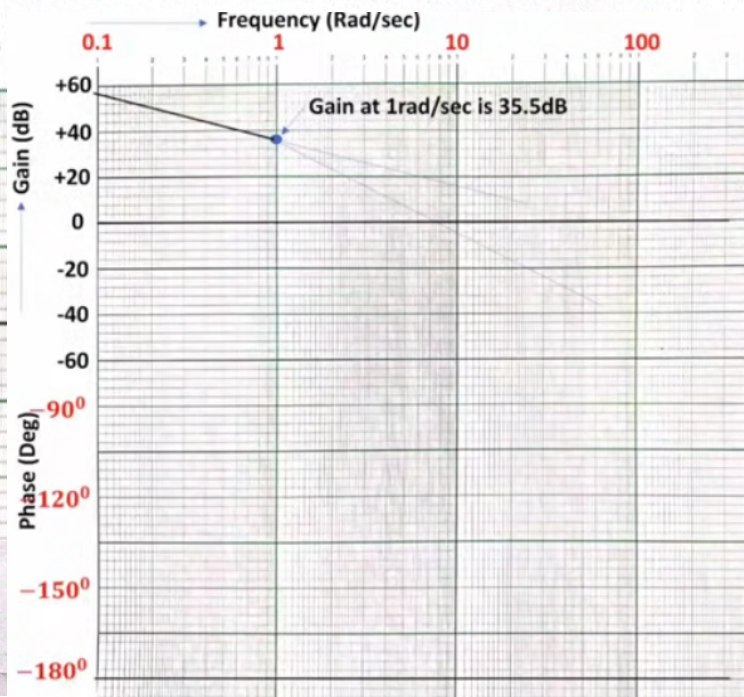
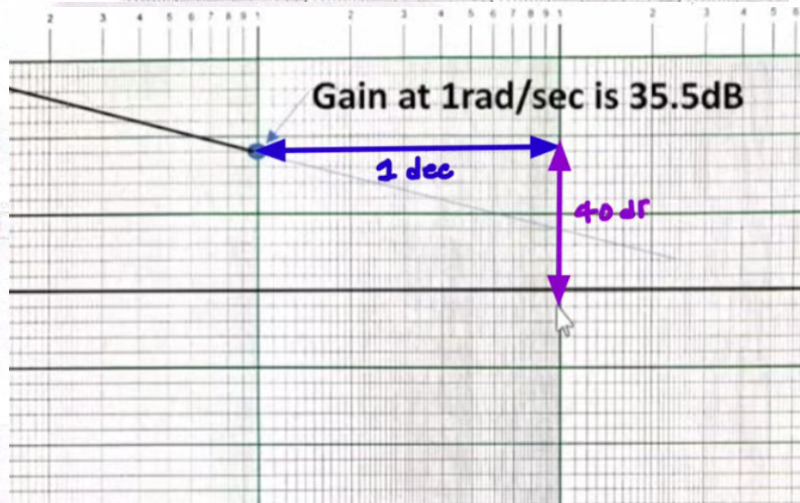
= -20 dB/dec

5

P

-20 dB/dec

= -40 dB/dec



Connen F.

P/Z

slope

Resultant slope

0

P

-20 dB/dec

= -20 dB/dec

1

P

-20 dB/dec

= -40 dB/dec

3

Z

+20 dB/dec

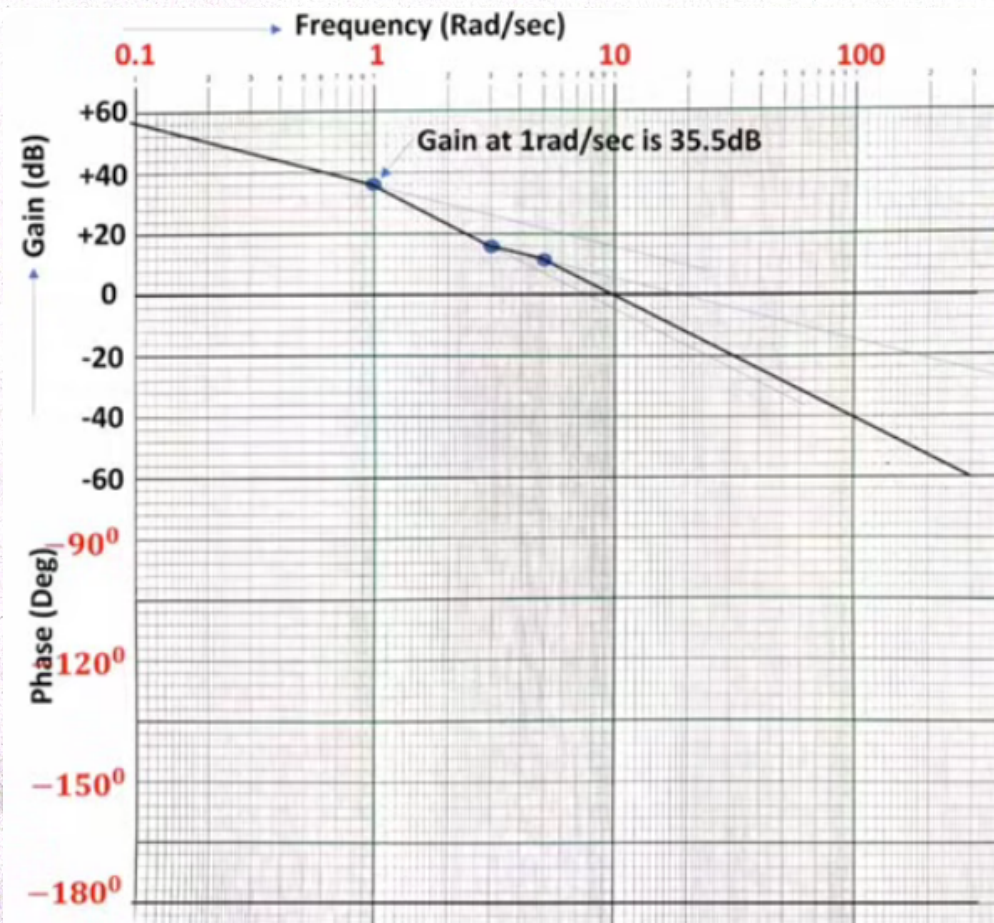
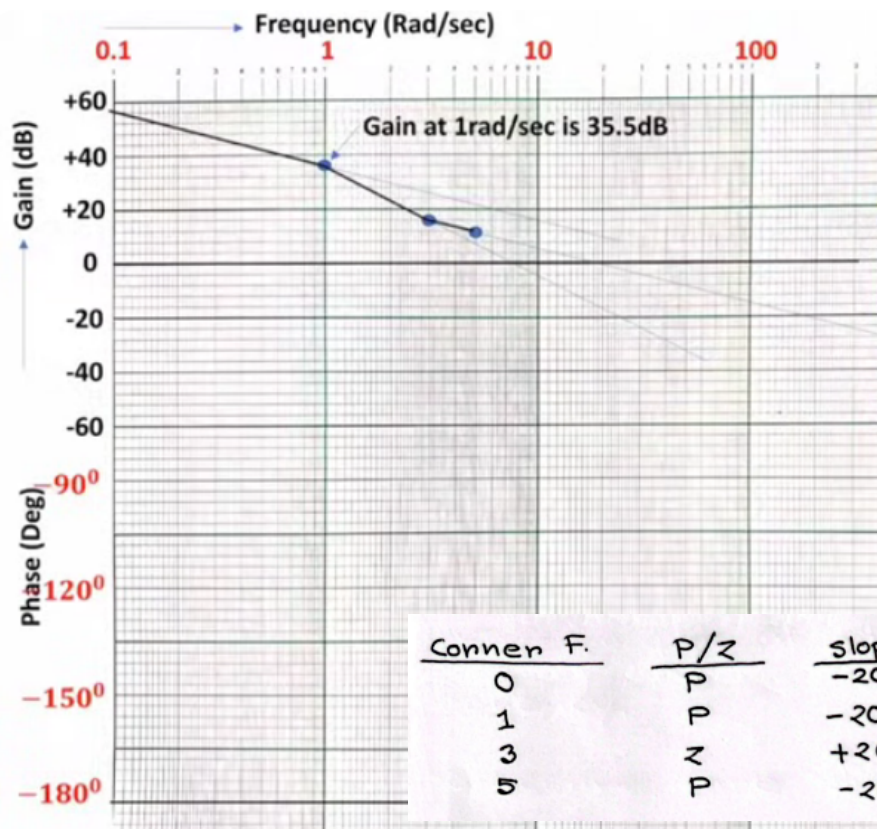
= -20 dB/dec

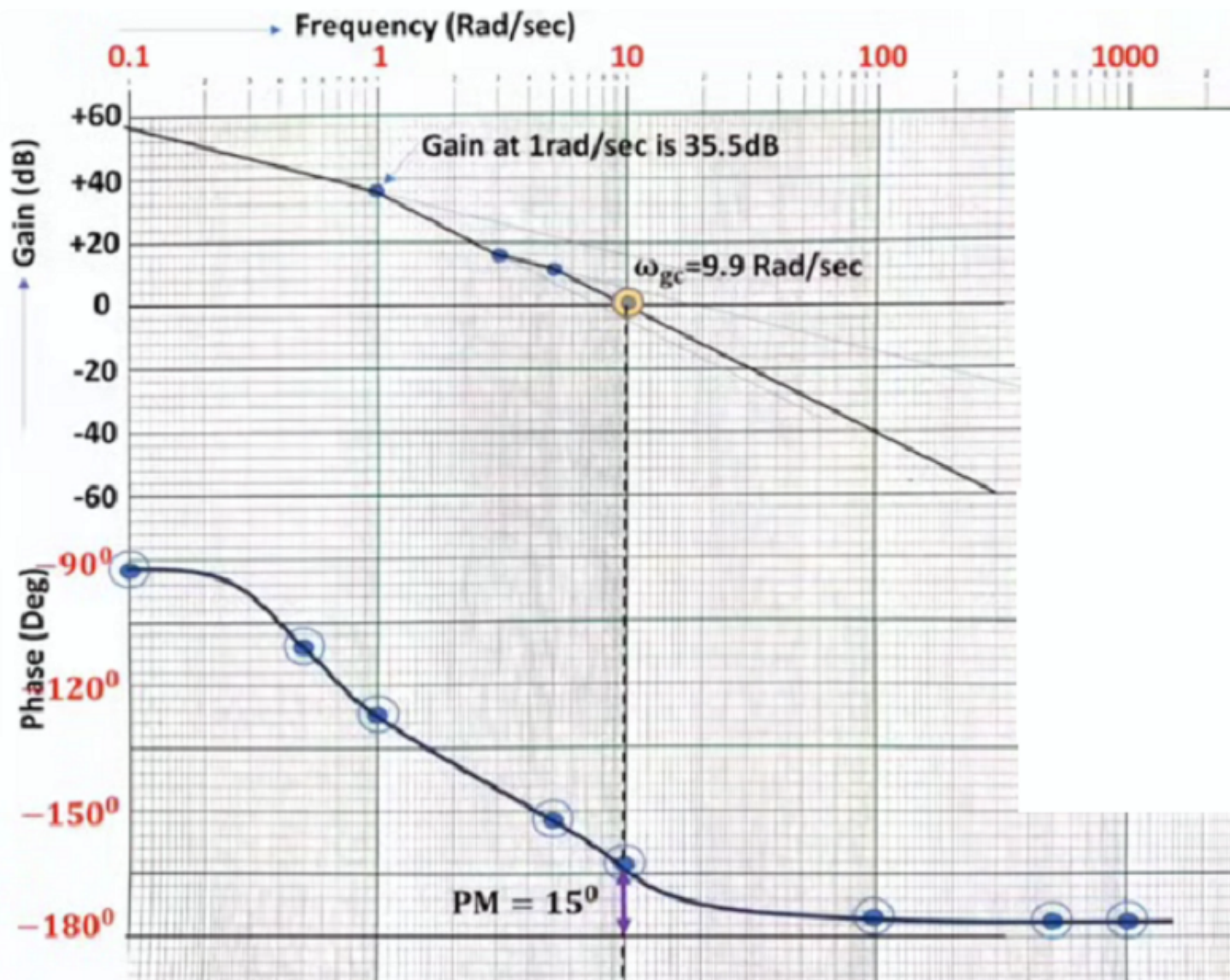
5

P

-20 dB/dec

= -40 dB/dec





$$\angle G(j\omega) = -90 - \tan^{-1}\omega - \tan^{-1}\frac{\omega}{5} + \tan^{-1}\frac{\omega}{3}$$

Make a table by taking some random frequencies.

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